# Executive Summary

## Factory-to-Customer Shipping Route Efficiency: Nassau Candy Distributor

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## Purpose

This study measures how efficiently Nassau Candy Distributor ships products from its five factories to customers in the United States and Canada. It identifies which routes are fast or slow, where delays concentrate, and how performance differs by region, state and shipping method. An interactive dashboard accompanies this summary.

## Business snapshot

| Orders / shipments | Customers | Total sales | Gross profit (margin) |
|---|---|---|---|
| 8,549 | 5,044 | $141,784 | $93,443 (65.9%) |

## Most important finding: the shipping dates need to be verified

Every record shows a shipment date **2.5 to 4.5 years after the order date** (904 to 1,642 days), and 95% of ship dates fall in the future. These cannot be real shipping times. The pattern suggests a systematic error in how ship dates were recorded or exported, although this cannot be proven from the data alone. **Until the source of the dates is confirmed, absolute delivery-time figures should not be quoted or used for decisions.**

To still learn something useful, the analysis removes the whole-year offset and compares routes with each other (a "relative" lead time). This is an assumption-based method for comparison only. It does recover the expected pattern between shipping methods, which gives some confidence that the ordering information in the data is real.

## What the data shows (relative comparison)

- **Shipping method matters most.** Same Day averages 0.4 relative days and Standard Class 5.3. Standard Class carries 60% of shipments, and 39% of them exceed a 5-day benchmark, compared with 3% of expedited shipments.
- **No factory or region is a confirmed bottleneck.** Apparent factory differences are largely explained by how many shipments each factory sends by Standard Class. The four customer regions are within about 0.1 day of each other.
- **Some specific routes deserve a look.** The highest-volume routes with above-average lead times include Lot's O' Nuts and Wicked Choccy's to New York, and both factories to Illinois. Alberta is slower than expected from both of the two largest factories, and Minnesota and New Mexico are also above average.
- **Differences between routes are real but small.** After allowing for shipping method, a typical route differs by about 0.4 days from expectation. Many routes have very few shipments, so the very slowest and fastest routes should be treated as leads, not conclusions.
- **Machine learning adds little beyond shipping method.** Predictive models were tested on customers they had never seen. None predicted lead time better than simply using the average for each shipping method, and models using only route information (factory, state, region) had essentially no predictive power. Predicting whether a shipment will be delayed improved only slightly (accuracy measure rose from 0.73 to 0.74). A route-level prediction tool is therefore not justified yet.

- **Costs of shipping are unknown.** The data holds product cost and profit but no freight cost, so the value of paying for faster shipping cannot be calculated. Profit margins are almost identical across shipping methods (about 66%).

## Recommendations

1. **Verify the ship-date source first.** Ask the data owner how ship dates are produced and obtain corrected dates.
2. **Collect missing data.** Record delivery dates, carrier, distance and freight cost per shipment so true transit performance and the cost of faster service can be measured.
3. **Review the highest-volume routes first** (New York and Illinois routes from the two large factories), since improvements there reach the most shipments.
4. **Put Alberta, Minnesota and New Mexico on a watch list** and check service arrangements for those destinations.
5. **Avoid factory-wide or region-wide programmes.** The evidence supports targeting specific routes, not whole factories or regions.
6. **Hold off on route-level prediction tools.** Use shipping method plus monitoring for now, and revisit predictive modelling once corrected dates and richer data are available.
7. **Re-run this analysis** once corrected dates are available; the dashboard recalculates automatically from a new file.

## What this study cannot say

It does not identify why any route is slow, because the data has no carrier, distance, weather or freight-cost information. Findings describe observed patterns only. Lead time here is order-to-dispatch time, not delivery time, and the relative comparison carries roughly one day of calendar noise. Full methods, tables and limitations are in the accompanying research paper.